

MIRPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY (MUST)



Department of Software Engineering

Lab no 03 — SQE

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C.1 Test Plan Identifier

Test Plan ID	LMS-TP-001
Version	1.0
Date	5th May 2026
Prepared by	University Library Management System
Reviewed by	Asim Javied
Status	Draft

C.2 Introduction

The University Library Management System (LMS) is a web-based application developed for a university to manage book loans, returns, student accounts, and library staff operations. The system serves two user roles: Students and Library Staff, interacting through a browser-based interface backed by a PostgreSQL database and Node.js server.

C.3 Test Items

Item	Version	Description
LMS (Frontend React.js)	V1.0.0	Browser-rendered React application served from https://lms-staging.uni.edu
LMS Backend (Node.js/Express.js)	V1.0.0	REST API server; endpoints documented in LMS-API-SPEC-v1.0.pdf
LMS (PostgreSQL)	V14.2	Relational database containing book, loan, student, and notification data
Automation Service	V1.0.0	JWT-based authentication module handling login, session, and password reset
Test Data Set	V1.0 (LMS-TESTDATA-001)	Pre-populated database with 200 books, 50 student accounts, 5 staff accounts

C.4 Features to be Tested

Feature	Requirements Covered	Test Priority
User login with valid credentials (email + password)	REQ-01	High
Rejection of invalid login attempts (wrong password, non-existent email)	REQ-01	High
Account lockout after exactly 3 consecutive failed attempts	REQ-02	High
Account lockout message display and admin unlock process	REQ-02	Medium
Password reset email request and delivery within 2 minutes	REQ-03	Medium
Password reset link validity and expiry (24-hour link life)	REQ-03	Medium

Session automatic expiry after 30 minutes of inactivity	REQ-04	High
Session timer reset upon any user interaction before 30 minutes	REQ-04	Medium
Book search by title keyword (partial and full match)	REQ-05	High
Book search by author name (partial and full match)	REQ-05	High
Book search by ISBN (exact match only)	REQ-05	High
Filter search results by genre category	REQ-06	Medium
Filter search results by availability status (Available / On Loan)	REQ-06	Medium
Search response time under 2 seconds with standard test data set	REQ-07	High
Pagination of results exceeding 20 items	REQ-08	Low

C.5 Features NOT to be Tested

Feature NOT Tested	Justification
M3: Loan Management (REQ-09 to REQ-12)	Out of scope for Week 3 test cycle. Covered in Test Plan LMS-TP-002 (Week 4 lab).
M4: Notifications (REQ-13 to REQ-16)	Out of scope for Week 3. Covered in Test Plan LMS-TP-003. SMS gateway not available in test environment.
Load/stress testing (>100 concurrent users)	No load testing tool configured in this lab environment. Planned for Week 10.
Browser compatibility testing (Safari, Firefox, Edge)	Chrome v120+ is the designated test browser. Cross-browser testing planned for Week 8.
Mobile/responsive design testing	System is desktop-only in v1.0. Mobile version planned for v2.0.
Database backup and recovery procedures	Infrastructure testing — out of scope for application testing. Covered by DevOps team.
API security penetration testing	Requires specialized tooling (Burp Suite). Planned for Week 12 Security Testing lab.

C.6 Testing Approach

Aspect	Approach for LMS Week 3 Test Cycle
Testing Level	System testing — the entire integrated application (not unit or integration tests)
Testing Type	Primarily functional black-box testing; performance testing for REQ-07 and REQ-16
Execution Method	Manual test execution — automated execution planned for Week 9 onward
Black-Box Techniques	Equivalence Partitioning (login input validation); Boundary Value Analysis (session timeout, response times)
Test Priority Order	High priority test cases executed first (REQ-01, REQ-02, REQ-04, REQ-07); Medium then Low

Defect Management	Defects logged in JIRA using the standard defect report template; severity assigned per defect classification guide
Re-testing	All defects fixed by development team will be re-tested within 24 hours of fix deployment
Regression Testing	After each defect fix: execute smoke test (TC-001 to TC-005) to verify no regression in core functionality
Test Data Management	Pre-loaded test database (LMS-TESTDATA-001); test data restored between test sessions using database snapshot

C.7 Pass/Fail Criteria

Level	Criterion	Pass Condition	Fail Condition
Test Case	Individual test case result	Actual result exactly matches the Expected Result documented in the test case.	Any deviation from the Expected Result, including partial matches or incorrect data.
Test Cycle (Exit)	Completion conditions for this test plan	100% of High priority test cases executed; 100% of Medium priority executed; 0 open Critical defects; 0 open High defects; test report reviewed and signed off.	Any open Critical defect; more than 10% of High priority test cases not executed; test report not reviewed within 5 business days.
Defect Severity — Critical	System crash or complete feature failure	N/A — critical defects immediately halt testing	Login module cannot be accessed; system crashes on startup; data corruption observed
Defect Severity — High	Major functional defect with no workaround	Feature works but with non-critical deviation	Account lockout not working; session does not expire; search returns no results
Defect Severity — Medium	Functional defect with available workaround	Minor deviation with documented workaround	Error message wording incorrect; pagination controls misaligned
Defect Severity — Low	Cosmetic or minor UI issue	Cosmetic defect does not affect function	Spelling error in UI text; minor layout issue on non-critical page

C.8 Suspension Criteria & Resumption Requirements

Suspension Criterion	Threshold	Action Required
Critical defect blocking all test cases	1 or more open Critical defects in the login module (REQ-01)	Halt all testing. Notify development lead immediately. Resume only after fix is deployed and verified.
High defect rate	More than 20% of test cases in current sprint are failing with High severity defects	Pause testing. Hold defect triage meeting within 24 hours. Resume after defect resolution plan agreed.

Test environment unavailable	LMS staging server unreachable for more than 2 hours during a test session	Suspend test session. Contact DevOps. Resume when environment is verified stable.
Test data corruption	Test database state deviates from LMS-TESTDATA-001 baseline	Halt testing. Restore database from snapshot. Re-verify test environment before resuming.
Build quality too low	More than 30% of smoke test cases (TC-001 to TC-005) fail on a new build	Reject the build. Do not proceed with further testing on this build. Request a new build from development.